

Mapping the Biodiversity in Duke Forest Utilizing Advanced Remote Sensing techniques

Biodiversity serves as a cornerstone for most ecosystem functions. Traditional methods for surveying forest biodiversity, such as forest inventories, are time-consuming and costly. This limits our ability to understand biodiversity on a large scale and monitor its dynamics. Advanced remote sensing techniques, such as imaging spectroscopy and Light Detecting And Ranging (LiDAR), offer a rapid and efficient alternative for biodiversity monitoring. Duke Forest, comprising various divisions with diverse species compositions, presents an excellent opportunity for conducting forest biodiversity research using advanced remote sensing technologies.

We are seeking students interested in forest biodiversity/fieldwork/spatial ecology/remote sensing to join this project. The project aims to conduct comprehensive biodiversity monitoring in Duke Forest using state-of-the-art remote sensing technologies, such as the Terrestrial Laser Scanning (TLS). Students will be expected to collect, process, and analyze LiDAR data acquired by the TLS. Training will be provided on the advanced RIEGL VZ-400i scanner, which will be used in the field to scan some selected sites within Duke Forest. Our data collection schedules will be determined based on both feasibility and team availability. Data processing will initially take place in Riegl Scan Pro and Tree analyzer software, for which manuals and operational guidance will be provided. After completing individual tree segmentation (ITS) tasks in the software, students will obtain LiDAR point cloud data for every individual tree at the study sites. Subsequent analyses will explore more complex aspects of forest structural information, such as niche space availability and forest gaps, and compare these with other data sources, such as ground sampling and airborne imagery. These analytical processes will involve coding in R. These analytical processes will involve coding in R. We foster an environment that encourages students to independently explore these techniques, with guidance and support available to facilitate their learning journey. Effective communication with students is crucial to address potential issues and explore research points of interest during this process. We look forward to receiving feedback from students to ensure a productive and insightful research experience.

Name of research group, project, or lab

Qiu Spatial Ecology and Environmental Data Science (SEEDS)

Website

www.tongqiulab.com

What is the context and setting for this research?

The Qiu Spatial Ecology and Environmental Data Science (SEEDS) lab is a dynamic group with Postdocs, Ph.D. students, and Master of Environmental Management (MEM) students. We aim to understand the causes and consequences of ecosystem function changes under global change. We use a highly interdisciplinary approach, including satellite and airborne remote sensing, field sampling, monitoring networks, and environmental data science. We are also committed to providing scientific evidence to federal and state agencies to inform conservation planning and policymaking.

Motivated undergraduate students who are eager to learn the interaction of remote sensing and ecology and their applications in environmental science are highly valued. Undergraduate students will get a chance to interact with members of the Qiu SEEDS lab at a weekly basis.

Representative publication**Logistics Information:****Project categories**

Environment

Student ranks applicable

Sophomore

Junior

Student qualifications

- Interest in trees, forest biodiversity, remote sensing, data analysis
- Organized, curious, meticulous, diligent
- Some experience with GIS or programming, in any language, would be useful

I expect a commitment of 5-7 hours per week for one semester but, if things go well, I would be happy to extend more opportunities. I have plenty of other research projects that would benefit from undergraduate student help.

Hours per week

5-7

Compensation

Paid - General

Paid - Work-Study Required

Unpaid - Ind. Study Credit

Unpaid - Volunteer

Number of openings

2

Techniques learned

- GIS
- Fieldwork in forest
- LiDAR data processing
- Geospatial programming

Project start

September 2024

Contact Information:

Mentor

[Johnny Ryan Mentor star](#)

[Expectations and Structure star](#)

jr555@duke.edu

[Faculty](#)

Name of project director or principal investigator

Tong Qiu

Email address of project director or principal investigator

tong.qiu@duke.edu

1 sp. | 0 appl.

Hours per week

5-7

Project categories

